



CASE HISTORY | PHARMACEUTICAL & BIOTECH

Cell Recycle Ceramic MF + UF for *Serratiopeptidase* Production

Broth Clarification, Enzyme Recovery & Concentration · Pharmaceutical & Biotech Sector

SITE INFORMATION

INDUSTRY

Pharmaceutical & Biotech

SYSTEM SUPPLIED

Ceramic MF followed by Polymeric UF (Two Stage)

MF CAPACITY

3,000 litres/hr

UF CAPACITY

3,000 litres/hr

UNITS SUPPLIED

Two such commercial units supplied

- >90–95% enzyme activity recovered in MF permeate
- 30–35x concentration by UF
- 97–99% Serratiopeptidase activity retained during UF concentration
- Running successfully for 4+ years
- Automatic advanced back-pulse design for steady-state flux

OBJECTIVE

Cell Recycle / Broth clarification — separating cells and biomass from the fermentation product (enzyme) by Microfiltration, followed by further concentration of the enzyme by Ultrafiltration (30–35x).

SCOPE

Ceramic MF + UF two-stage semi-automatic PLC system. Complete automation for process and CIP cycles, including in-process back-pulse.

TECHNOLOGIES

- Ceramic Microfiltration (MF)
- Polymeric Ultrafiltration (UF)
- Diafiltration (integrated in MF stage)
- PLC-based automation
- In-process back-pulse design

>90–95% Enzyme Activity Recovered · 30–35x UF Concentration · 97–99% Activity Retained · 4+ Years Running

INTRODUCTION

Serratiopeptidase is a proteolytic enzyme with fibrinolytic and anti-inflammatory activity, commonly used as a drug for pain and inflammation. The bacterial broth from fermentation typically has a Packed Cell Volume (PCV) of 25–35%, with active enzyme having an activity of 12,000–15,000 SPU/ml, along with other soluble micro and macro nutrients added during the fermentation process.

Permionics was engaged to provide a two-stage membrane system — Ceramic Microfiltration followed by Polymeric Ultrafiltration — to separate cells from the enzyme-rich permeate and subsequently concentrate the enzyme to levels required for downstream processing. Two such commercial units have been supplied and have been running at production scale for over four years.

THE CHALLENGE

Feed Characteristics. Serratiopeptidase bacterial broth presents a challenging feed: a high PCV of 25–35%, active enzyme at 12,000–15,000 SPU/ml, and a complex mixture of soluble micro and macro nutrients from the fermentation process. The downstream requirement was to recycle the bacterial cells and recover a cell-free, clear permeate carrying high enzyme activity — while concentrating that permeate 30–35x before the next downstream step.

Process Requirement. At downstream post-fermentation, the need was to recycle the bacterial cell and recover cell-free clear permeate with more than 90–95% enzyme activity, then concentrate the MF permeate to enrich Serratiopeptidase activity to 1.5 to 2.5 lacs SPU/ml before the next step.

Parameter	Value
Broth PCV	25–35%
Active Enzyme Activity	12,000–15,000 SPU/ml
MF Capacity	3,000 litres/hr
UF Capacity	3,000 litres/hr
Target UF Concentration	30–35x fold
Target Enzyme Activity (post UF)	1.5 to 2.5 lacs SPU/ml

Stage 1 — Ceramic Microfiltration (MF) System & Process

- ✓ At downstream post fermentation, the need was to recycle bacterial cells and recover cell-free clear permeate with >90–95% enzyme activity. This has been achieved successfully via MF Concentration + Diafiltration process, running since the last 4 years.
- ✓ Automatic advanced back-pulse design provision helps maintain steady-state flux during entire operation, completing the batch within stipulated time.

The MF stage uses ceramic membranes across 3 loop areas totalling approximately 120 m² of membrane area, enabling high-throughput clarification of the dense bacterial broth.



Cell Recycle MF system (3 loops, ~120 m² membrane area) — Ceramic Microfiltration installation at production site.

MF STAGE RESULTS

- Cell-free clear permeate recovered with >90–95% enzyme activity
- MF Concentration + Diafiltration process running successfully for 4+ years
- Steady-state flux maintained throughout batch via back-pulse design

UF STAGE RESULTS

- MF permeate concentrated 30–35 fold by UF
- Serratiopeptidase* activity enriched to 1.5–2.5 lacs SPU/ml
- 97–99% *Serratiopeptidase* activity retained during concentration step
- UF also removes undesired organic/inorganic components — improving purity in the same step
- Running since last 4 years at production scale

VALUE DELIVERED

- Single integrated two-stage system delivers clarification, concentration, and purity improvement
- Eliminates need for separate unit operations for each function
- PLC automation with CIP cycles enables consistent, low-intervention operation
- 4+ years of continuous commercial-scale reliability

FOR MORE INFORMATION

Permionics Membranes Pvt. Ltd.

www.permionics.com

Pharmaceutical & Biotech
Division
India · Global

Stage 2 — Polymeric Ultrafiltration (UF) System & Process

- During the Microfiltration process, the enzyme permeates through the membrane leaving behind bacterial debris and cells. The Diafiltration process dilutes the enzyme activity to approximately 5,000–8,000 SPU/ml, making it important to enrich and purify before the next downstream step.
- UF process helps to concentrate the clear MF permeate 30–35 fold, enriching *Serratiopeptidase* activity to 1.5 to 2.5 lacs SPU/ml. The UF system running since the last 4 years retains 97–99% *Serratiopeptidase* activity during the concentration step.
- While retaining enzymes during concentration, UF also removes undesired organic and inorganic components. Thus, in the same process, apart from concentration, UF also improves the purity of the *Serratiopeptidase*.



UF system (720 m² membrane area) — Polymeric Ultrafiltration installation at production site.

OPERATIONAL RESULTS

>90–95%

ENZYME ACTIVITY
RECOVERED IN MF
PERMEATE

30–35x

CONCENTRATION
FACTOR BY UF

97–99%

SERRATIOPEPTIDASE
ACTIVITY RETAINED IN UF

4+ Yrs

CONTINUOUS COMMERCIAL
SCALE OPERATION

Performance Parameter	Stage	Result
Cell separation from broth	MF	Cell-free clear permeate recovered with >90–95% enzyme activity
Enzyme activity in MF permeate post Diafiltration	MF	~5,000–8,000 SPU/ml (diluted; enriched in UF stage)
UF concentration factor	UF	30–35x fold
<i>Serratiopeptidase</i> activity post UF	UF	1.5 to 2.5 lacs SPU/ml
<i>Serratiopeptidase</i> activity retention in UF	UF	97–99%
Additional benefit of UF	UF	Removes undesired organic/inorganic components — improves purity in same step
System operational track record	Both stages	Running continuously for 4+ years at commercial production scale

CONCLUSIONS

Permionics' integrated two-stage Ceramic MF + Polymeric UF system has delivered consistent, high-performance *Serratiopeptidase* recovery and concentration at commercial production scale for over four years. The MF stage achieves greater than 90–95% enzyme activity recovery in a cell-free permeate via Concentration + Diafiltration, while the back-pulse design sustains steady-state flux through the batch. The UF stage then concentrates this permeate 30–35 fold — enriching activity to 1.5 to 2.5 lacs SPU/ml while retaining 97–99% of *Serratiopeptidase* activity and simultaneously removing undesired organic and inorganic components, improving product purity within the same unit operation. Two such commercial units have been supplied and continue to run reliably, demonstrating Permionics' capability to deliver integrated, multi-stage membrane solutions for complex pharmaceutical bioprocess separations.